VCU Discrete Mathematics Seminar

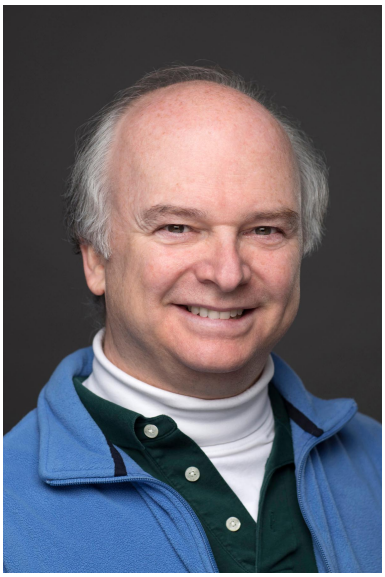
Pebbling Cartesian Products of C_5

Prof David Herscovici
(Quinnipiac University)

Thursday, Feb. 5 (**Special Day!**)
1:00-1:50 EST

On Zoom, and watching in 4145 Harris:

`https://vcu.zoom.us/j/81475528886`
`password=graphs2357`



We imagine a distribution of pebbles on the vertices of a graph. Chung defined a *pebbling move* as the removal of two pebbles from some vertex and the addition of a pebble to an adjacent vertex. Then the *pebbling number* of a graph G is the smallest number $\pi(G)$ such that every distribution of $\pi(G)$ pebbles allows one pebble to reach any specified, but arbitrary vertex.

We discuss the basics ideas of pebbling and the pebbling number $\pi(G)$ of a connected graph G. We then define the Cartesian product $G \square H$ of two graphs, and explain Graham's conjecture that $\pi(G \square H) \leqslant \pi(G)\pi(H)$. Finally, we sketch a proof that the pebbling number of $C_5 \square G$ is at most $\pi(C_5 \square G) \leqslant 5\pi(G)$ for any graph G that satisfies the two-pebbling property.

For the DM seminar schedule, see:

`https://go.vcu.edu/discrete`